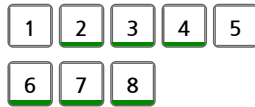


Quiz navigation



Finish review

Course Home

Course Info ▴

Syllabus

Participants list

Grade/Attendance ▴

Offline-Attendance

Grades

Students Notifications ▾

Others ▾

Activities/Resources +



Discrete Mathematics

11Week [20 April - 26 April]

Quiz 11

Started on	2024-05-01 20:31
State	Finished
Completed on	2024-05-01 23:43
Time taken	3 hours 11 mins
Grade	13.73 out of 15.00 (92%)

Question 1

Partially correct

Mark 1.33 out of 2.00

The relation

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$$

on $A = \{1, 2, 3, 4\}$ is

Select one or more:

- ☒ reflexive ✓
- ☐ symmetric
- ☒ antisymmetric ✓
- ☐ transitive

Your answer is partially correct.

You have correctly selected 2.

The correct answers are: reflexive, antisymmetric, transitive

Question 2

Correct

Mark 1.00 out of 1.00

Find reflexive closure of the relation

$$R = \{(b, c), (b, e), (c, e), (d, a), (e, b), (e, c)\}$$
 on

$$A = \{a, b, c, d, e\}.$$

From the list below choose the pairs which you add to R , when construct the closure.

Select one or more:

☒ (a, a) ✓☐ (a, b) ☐ (a, c) ☐ (a, d) ☐ (b, a) ☒ (b, b) ✓☐ (b, d) ☐ (c, a) ☐ (c, b) ☒ (c, c) ✓☐ (c, d) ☐ (d, b) ☐ (d, c) ☒ (d, d) ✓☐ (d, e) ☐ (e, a) ☐ (e, d) ☒ (e, e) ✓

Your answer is correct.

The correct answers are: (a, a) , (b, b) , (c, c) , (d, d) . (e, e)

Question 3

Correct

Mark 1.00 out of 1.00

Find symmetric closure of the relation

$$R = \{(b, c), (b, e), (c, e), (d, a), (e, b), (e, c)\}$$
 on

$$A = \{a, b, c, d, e\}.$$

From the list below choose the pairs which you add to R , when construct the closure.

Select one or more:

- ☐ (a, a)
- ☐ (a, b)
- ☐ (a, c)
- ☒ (a, d) ✓
- ☐ (b, a)
- ☐ (b, b)
- ☐ (b, d)
- ☐ (c, a)
- ☒ (c, b) ✓
- ☐ (c, c)
- ☐ (c, d)
- ☐ (d, b)
- ☐ (d, c)
- ☐ (d, d)
- ☐ (d, e)
- ☐ (e, a)
- ☐ (e, d)
- ☐ (e, e)

Your answer is correct.

The correct answers are: (a, d) , (c, b)

Question 4

Correct

Mark 1.00 out of 1.00

Find transitive closure of the relation

$$R = \{(b, c), (b, e), (c, e), (d, a), (e, b), (e, c)\}$$
 on

$$A = \{a, b, c, d, e\}.$$

From the list below choose the pairs which you add to R , when construct the closure.

Select one or more:

- ☐ (a, a)
- ☐ (a, b)
- ☐ (a, c)
- ☐ (a, d)
- ☐ (b, a)
- ☒ (b, b) ✓
- ☐ (b, d)
- ☐ (c, a)
- ☒ (c, b) ✓
- ☒ (c, c) ✓
- ☐ (c, d)
- ☐ (d, b)
- ☐ (d, c)
- ☐ (d, d)
- ☐ (d, e)
- ☐ (e, a)
- ☐ (e, d)
- ☒ (e, e) ✓

Your answer is correct.

The correct answers are: (b, b) , (c, b) , (c, c) , (e, e)

Question 5

Partially correct

Mark 2.40 out of 3.00

Find the smallest equivalence relation R which contains the relation

$\{(a, b), (b, d), (c, e)\}$ on $A = \{a, b, c, d, e\}$.

From the list below choose all pairs which you add to the relation to construct R .

Select one or more:

- ☒ (a, a) ✓
- ☐ (a, c)
- ☐ (a, d)
- ☐ (a, e)
- ☒ (b, a) ✓
- ☒ (b, b) ✓
- ☐ (b, c)
- ☐ (c, a)
- ☐ (c, b)
- ☒ (c, c) ✓
- ☐ (c, d)
- ☒ (d, b) ✓
- ☐ (d, a)
- ☒ (d, d) ✓
- ☐ (d, e)
- ☒ (e, c) ✓
- ☐ (e, d)
- ☒ (e, e) ✓

Your answer is partially correct.

You have correctly selected 8.

The correct answers are: (a, a) , (a, d) , (b, a) , (b, b) , (c, c) , (d, b) , (d, a) , (d, d) , (e, c) , (e, e) .

Question 6

Correct

Mark 2.00 out of 2.00

If

$$R = \{(a, a), (a, b), (a, d), (b, b), (b, a), (b, d), (c, c), (c, e), (d, d), (d, a), (d, b), (e, c), (e, e)\}$$

, find

$$[a]_R = \{ a, b, d \}$$

$$[b]_R = \{ a, b, d \}$$

$$[c]_R = \{ c, e \}$$

$$[d]_R = \{ a, b, d \}$$

$$[e]_R = \{ c, e \}$$

Question 7

Complete

Mark 2.00 out of 2.00

What is the partition of the set $A = \{a, b, c, d, e\}$ formed by the equivalence relation

$$R = \{(a, a), (a, b), (a, d), (b, b), (b, a), (b, d), (c, c), (c, e), (d, d), (d, a), (d, b), (e, c), (e, e)\}$$

?

Draw it and attach it below.

 U2210251.jpg

Comment:

Question 8

Complete

Mark 3.00 out of 3.00

A development project requires the completion of seven tasks

A, B, C, D, E, F, G . Some of these tasks can be started only

after other tasks are finished. Set $X \preceq Y$ if task Y cannot be started until task X has been completed. This relation is given as

follows $A, C \preceq B$, $B, E \preceq F$, $B \preceq D$ and

$D, F \preceq G$. Find an order in which these tasks can be carried out

to complete the project, if B cannot be right after C and G cannot be right after D .

One answer is enough.

ACEBDFG

Comment: